Name: Bushrat M Pathan

USN: 01SU24CS030

Q1: Variance and Bias (Diagram, overfit, underfit) - For best fit model should we have low bias or high variance, low bias or low variance, high bias or high variance, high bias or low variance.

Variance and Bias in Machine Learning

Introduction

In Machine Learning, the performance of a model depends largely on two important concepts: **Bias** and **Variance**. These two factors determine whether a model will generalize well to new data or not. Understanding bias and variance helps in selecting the best model and avoiding common problems such as underfitting and overfitting.

The main goal of any machine learning model is to achieve a balance between bias and variance so that it performs well on both training data and unseen test data.

1. Bias

Bias refers to the error introduced by approximating a real-world problem with a simplified model. It measures how far the model's predictions are from the actual values.

A model with **high bias** makes strong assumptions about the data. Because of this, it may fail to capture important patterns.

Characteristics of High Bias:

- Model is too simple.
- Ignores complex relationships.
- Performs poorly on both training and testing data.
- Leads to **Underfitting**.

Example:
Using a straight line (linear model) to fit data that actually follows a curved pattern.

2. Variance

Variance refers to how much the model's predictions change when trained on different datasets. It measures the model's sensitivity to fluctuations in the training data.

A model with high variance learns too much from the training data, including noise and random fluctuations.

Characteristics of High Variance:

- Model is too complex.
- Learns noise in the data.
- Performs very well on training data.
- Performs poorly on test data.
- Leads to Overfitting.

Example:
Using a very high-degree polynomial curve that passes through every data point.

<u>Underfitting (High Bias, Low Variance)</u>

Underfitting occurs when the model is too simple to capture the underlying pattern of the data.

Combination:

High Bias + Low Variance

Explanation:

- The model cannot learn properly.
- It performs poorly on training data.
- It also performs poorly on new data.

In this case, the model has high error because it has not captured enough information from the dataset.

Overfitting (Low Bias, High Variance)

Overfitting occurs when the model is too complex and memorizes the training data.

Combination:

Low Bias + High Variance

Explanation:

- The model performs very well on training data.
- But it performs poorly on testing data.
- It captures noise instead of real patterns.

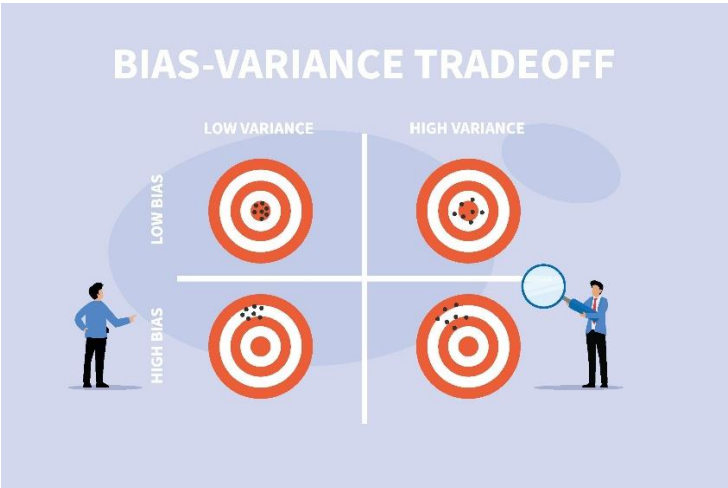
In this situation, the model does not generalize well to new data.

Bias–Variance Tradeoff

There is always a tradeoff between bias and variance.

- If we reduce bias too much, variance may increase.
- If we reduce variance too much, bias may increase.

The objective is to find a balance between the two so that the model achieves optimal performance.



Best Fit Model (Low Bias, Low Variance)

A Best Fit Model is the ideal machine learning model that properly captures the true pattern of the data without being too simple or too complex. It maintains a balance between bias and variance, allowing it to perform well on both training data and unseen test data.

Combination:

Low Bias + Low Variance

Explanation:

- The model is flexible enough to learn the real relationship between input and output (low bias).
- It does not memorize noise or random fluctuations in the training data (low variance).
- It generalizes well to new data.
- Both training error and testing error are low.
- The model captures the overall trend of the data without unnecessary complexity.

In graphical terms, the best fit curve smoothly follows the general pattern of the data points. It does not underfit like a straight line, and it does not overfit like a highly wavy curve that passes through every point.

A best fit model ensures reliable predictions, stable performance, and high accuracy. Achieving this balance is known as the **Bias–Variance Trade off**, and it is the main objective in building effective machine learning models.

Conclusion

The best fit model in machine learning is achieved by balancing bias and variance. It combines low bias and low variance to provide accurate and generalizable predictions. Instead of being too simple or too complex, the model maintains optimal complexity to capture real patterns in the data. Achieving this balance is essential for building effective, reliable, and high-performing AI systems.

Q2: Take one Domain and draw the graph (Normal distribution) (Empirical rule)

Introduction

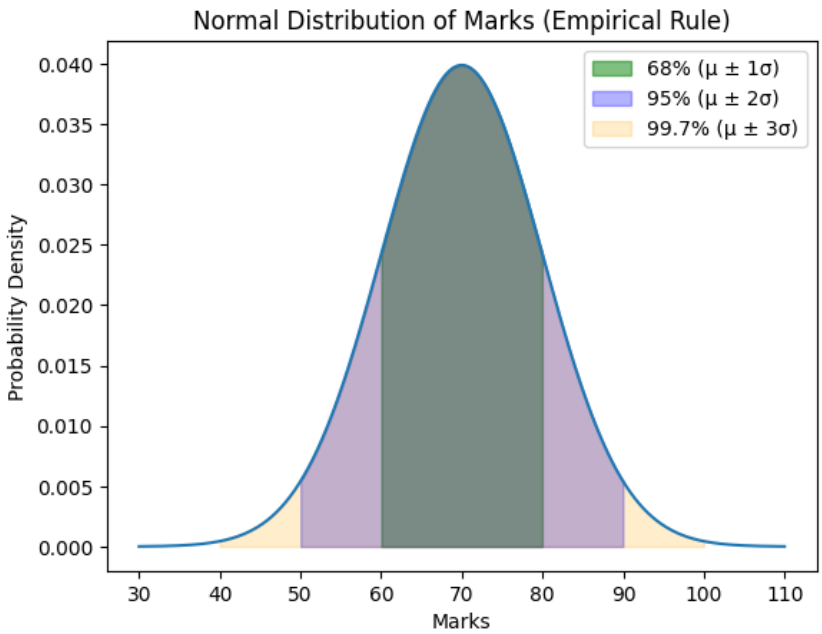
The Normal Distribution, also known as the Gaussian Distribution, is one of the most important probability distributions in statistics. It is a symmetric, bell-shaped curve that represents how data values are distributed around the mean (average). Many real-world phenomena such as exam scores, heights, IQ levels, and measurement errors follow a normal distribution.

One important concept related to normal distribution is the Empirical Rule (68–95–99.7 Rule). This rule states that in a normal distribution:

- Approximately 68% of the data falls within 1 standard deviation (σ) from the mean (μ).
- Approximately 95% of the data falls within 2 standard deviations.
- Approximately 99.7% of the data falls within 3 standard deviations.

In this example, we consider the domain of Student Exam Scores to illustrate the normal distribution and empirical rule.

Here is the Normal Distribution graph for the domain: Student Exam Scores with:



- Mean (μ) = 70
- Standard Deviation (σ) = 10

- Vertical lines at:
  - μ
  - $\mu \pm 1\sigma$

    It means: $70 \pm 1(10)$

    So:

    $70 - 10 = 60$

    $70 + 10 = 80$

    $\mu \pm 1\sigma = 60$ to 80
  - $\mu \pm 2\sigma$

    It means: $70 \pm 2(10)$
    So:

    $70 - 20 = 50$

    $70 + 20 = 90$

    $\mu \pm 2\sigma = 50$ to 90
  - $\mu \pm 3\sigma$

    It means: $70 \pm 3(10)$

    So:

    $70 - 30 = 40$

    $70 + 30 = 100$

    $\mu \pm 3\sigma = 40$ to 100

Empirical Rule Explanation:

- About 68% of students score between 60 and 80 ($\mu \pm 1\sigma$)
- About 95% score between 50 and 90 ($\mu \pm 2\sigma$)
- About 99.7% score between 40 and 100 ($\mu \pm 3\sigma$)
- This is called the **68–95–99.7 Rule (Empirical Rule)**.

Q3: If $\mu = 55$, $\sigma_{4a} = 4$, $\sigma_{4\beta} = 10$, $\sigma_{4c} = 15$, In this which is better

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Given:

- Mean (μ) = 55
- Standard deviation of A (σ_a) = 4
- Standard deviation of B (σ_β) = 10
- Standard deviation of C (σc) = 15

All three distributions have the **same mean (55)**.

So the only difference is how **spread out** the data is.

Standard deviation (σ) tells us how much values vary from the mean.

- Smaller σ → Data is more consistent (less spread)
- Larger σ → Data is more spread out (less consistent)

Comparison:

Distribution	Mean (μ)	Standard Deviation (σ)	Interpretation
A	55	4	Very less spread (more consistent)
B	55	10	Moderate spread
C	55	15	Highly spread (less consistent)

Which is Better?

Distribution A ($\sigma = 4$) is better

because:

- It has the smallest standard deviation
- Data values are closer to the mean (55)
- It is more stable and consistent
- Less variability means more reliability

If all means are equal ($\mu = 55$), the distribution with the **lowest standard deviation ($\sigma = 4$)** is considered better.

Distribution A is the best.

Q4: Take One Domain and Build Business Understanding

Domain: E-Commerce (Online Shopping Platform)

1. Introduction

Business understanding is the first and most important phase in any data-driven or AI project. It involves clearly defining the business problem, understanding objectives, identifying stakeholders, and determining how data and analytics can provide solutions. Without proper business understanding, even the most advanced technical solution may fail to deliver value.

In this case, we select the **E-commerce domain** (online shopping platforms like Amazon or Flipkart) to build business understanding.

2. Business Problem

One of the major challenges faced by e-commerce companies is improving sales conversion rates while maintaining customer satisfaction.

An e-commerce company wants to:

- Increase sales revenue
- Improve customer satisfaction
- Reduce cart abandonment
- Provide personalized recommendations

The main problem is that many customers browse products but do not complete purchases. The company wants to understand customer behaviour and improve conversion rates.

3. Business Objectives

The key objectives of the company are:

- Increase overall sales by 15%
- Improve customer retention
- Provide personalized shopping experience
- Optimize inventory management

These objectives help guide data analysis and AI model development.

4. Stakeholders

Stakeholders are people who are involved in or affected by the project:

- Business managers
- Marketing team
- Data analysts
- IT department
- Customers

Each stakeholder has different expectations. For example, marketing wants better targeting, while management wants higher profits.

5. Key Business Questions

To understand the business clearly, we must ask:

- Why are customers abandoning carts?
- Which products are most popular?
- Which customers are likely to purchase again?
- How can we personalize product recommendations?
- What factors influence customer buying decisions?

6. Data Requirements

To solve these questions, the company needs:

- Customer data (age, location, purchase history)
- Product data (price, category, ratings)
- Transaction data
- Website interaction data (clicks, time spent)

7. Success Criteria

The project will be considered successful if:

- Conversion rate increases
- Customer satisfaction improves

- Sales revenue grows
- Repeat purchases increase

These measurable indicators help evaluate performance.

Conclusion

Business understanding in the e-commerce domain involves identifying the main business problem, setting clear objectives, understanding stakeholders, and defining measurable success criteria. By properly understanding business needs, data-driven solutions such as recommendation systems and predictive analytics can be effectively developed to improve sales and customer experience.